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Recent advancements in revenue management of taxi services: a systematic review and research agenda

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Abstract
This study aims to present a systematic review of the most recent literature on revenue management practices in taxi services and illustrate insightful future directions for research. The classical works of M. K. Geraghty and W. J. Carroll have served as a bedrock for understanding the revenue management of taxi services. However, the advent of online platforms and advances in technology over the last 20 years has led to a wide range of studies that adapt to improve the profitability of taxi services. We reviewed the literature on taxi services' revenue management practices by collecting the most relevant research articles based on specified delimiting criteria. A total of 110 research articles were analyzed to gain vital insights into revenue management practices in taxi services. We study the revenue management framework from Pricing, Market structure, Business Models and technology, Structure, and Marketing viewpoints. The trends emerging from the literature provide a thorough understanding of the research directions of this domain, along with future scopes of research in this area. Also, we explicitly identify research gaps that are relevant to academic research and practice.

Keywords Taxi services · Revenue management · Taxi pricing · Business models · Taxi market regulation · Shared mobility

JEL Classification R480 · D440 · D70 · L910 · R41

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1 Introduction

For several years, taxi services are an essential aspect of the urban transportation system. Taxi services can be divided into two, viz.—traditional taxi services, where drivers look for passengers by cruising on the streets, and platform-based taxi services, where drivers receive orders from the platform's application. Taxi services act as both a supplement and complement to public transportation (Austin and Zegras 2012; Hall et al. 2018). However, traditional taxi services occupy road space, aggravates traffic congestion, and cause pollution, particularly when searching for road-hailing passengers (Wong et al. 2015; Yang et al. 2014). The supply and demand mismatch in traditional taxi services concerning time and space leads to poor efficiency, which in turn causes passenger dissatisfaction (Nie 2017). On the other hand, the prolific developments in telecommunication technologies have led to many platform-based taxi services, such as Uber, Ola Cabs, Didi KuaiDi-Taxi, Grab, Go-Jek, Careem, Taxify, and Easy Taxi. Gone are the days, where passengers have to wait on the streets for hailing taxis. Likewise, the driver shall no longer spend time searching for passengers. The platform-based taxis connect passengers and drivers instantaneously by reducing the spatial barrier and maximizes taxi market efficiency (Wang and Yang 2019). However, the increased demand for taxi services leads to an exponential increase in traffic congestion and emissions (Çetin and Eryigit 2011).

On one hand, taxi services constantly focus on multi-fold objectives that include maximization of: revenue, market share, social welfare, and matching efficiency. On the other hand, taxi services also focus on minimizing: passenger waiting times, vacant driver hours, and passenger refunds. However, the taxi services are prone to significant losses in terms of revenue (Hawkins 2020). The majority of revenue gained by taxi services goes to the drivers as wage, and the rest to promotional campaigns, passenger refunds, and governmental taxes (Wang and Yang 2019).

Platform-based taxi services continuously face the challenge of supply–demand imbalance during the peak and non-peak hours (Cho et al. 2021). For many years, the traditional taxi services have used a low-pricing scheme during low-demand periods. At the time of low demand, the supply of taxi services is high, which lead to the reduction of taxi fares. However, lowering the taxi fare resulted in substantial losses for taxi service providers. Also, delay in implementation of technology was one of the key driving factors that lead to the taxi market crisis of taxi companies like National Car Rental and Hertz Corporation (Carroll and Grimes 1995; Geraghty and Johnson 1997). It was the airline industry that pioneered revenue management practices in the service sector. The airline industry targeted leisure passengers by selling seats at a reduced price to passengers willing to purchase an airline ticket with penalty schemes on cancellations and no-shows (Talluri and Van Ryzin 2004; Sierag et al. 2015). In the 1990s, National Car Rental and Hertz Corporation implemented revenue management practices that controlled the price, booking requests, and availability of taxis. Moreover, both the companies focused on leisure passengers to increase revenue, invested

in the technology for tracking the taxi fleets, and developed mathematical models for forecasting and pricing the rides optimally. The revenue management practices led to a substantial increase in revenues of both the companies and saved them from facing liquidation (Geraghty and Johnson 1997; Carroll and Grimes 1995). In addition to revenue maximization, the practitioners of revenue management focus on optimal use of existing facilities and modern technologies to increase network efficiency and vehicle safety, and reduce the human travel time (Ma 2021; Sun andTeichert 2022).

Traditional taxi services primarily focus on corporate customers using a fixed pricing strategy across most locations at a given time (Chen et al. 2020; Hörl et al. 2021). Generally, corporate passengers plan their travel on weekdays, and therefore the fleets were idle on weekends. Moreover, there was no penalty charged to passengers for cancellations (Carroll and Grimes 1995; Geraghty and Johnson 1997). Hence, passenger ride cancellations and the poor utilization rate of taxis were the most significant bottlenecks faced by traditional taxi services before the advent of online taxi platforms. It is essential to understand how revenue management contributes to the overall development of the taxi service providers and how its systematic use is vital for the future prospects of taxi services, which are currently struggling with profitability. Revenue management integrates information technologies with sophisticated modeling techniques for pricing the taxi rides, forecasting demand and fleet, and improving the service quality.

Researchers and practitioners in the service industry, viz.—airlines, hotels, and rental cars, view revenue management as a vital tool for their future prospects (Touraine 2021). Moreover, the expected revenue growth through revenue management capabilities is driving taxi services to acquire new technologies (Garvey 1997). Early development of mathematical models in revenue management in the service sector focused mainly on demand forecasting and revenue maximization (Littlewood 1972). Furthermore, revenue management models can be effectively used to price the taxi rides optimally, match demand and supply efficiently, maximize revenue or market share.

Forecasting the taxi demand is crucial since it can increase profits, help in the allocation of taxis, and improve the service quality (Qian et al. 2020). Forecasting taxi demand by using linear and nonlinear models has been in practice for a long time. Techniques like simple least square, recursive least square, Winter and Holt's model, neural network, support vector machines, and the combinations of them are well exploited (Cheu et al. 2006; Kek et al. 2005). The nonlinear models outperform the linear models, indicating that taxi fleets' demand is nonlinear in nature (Xu and Lim 2007). Further, a hybrid approach of using genetic and back-propagation algorithms is proven to find optimal demand forecasting solutions (McInerney and Dhawan 1993). Due to factors such as complexity and stochastic nature, accurate demand prediction is still a major challenge for taxi services (Zhao et al. 2020). Also, taxi services have potential to explore forecasting techniques in the areas of overbooking, cancellation and no-shows (Belobaba 1987; Smith et al. 1992; VINOD 1995).

In addition to demand prediction, fleet forecasting helps taxi services to reduce operating costs and improve service levels (Dyshkantiuk et al. 2020). Also, fleet

forecasts enable taxi services to efficiently plan their vehicles to the locations with high taxi demand (Cheu et al. 2006). The technology that gained a lot of attention for taxi services is a taxi probe system, which provides the taxi's historical data using compact sensors that collect the vehicle information (Mukai and Yoden 2012). Typically, the latitude and longitude data of customer pickup are stored in the taxi probe system. Also, the taxi probe system helps calculate advanced information such as identifying traffic congestion, measuring travel time.

Figure 1 shows the revenue management framework for taxi services. In this study, we examine revenue management from the viewpoints of pricing, market structure, business models, and structural aspects. Rest of the paper is described as follows: Sect. 2 demonstrates the methodology for the literature study. Section 3 represents the pricing component of the revenue management framework, followed by the market structure of taxi services in Sect. 4. Next, we discuss the business models in Sect. 5. The structural elements are discussed in Sect. 6. Finally, we discuss the research gaps in Sect. 8, and we conclude the paper in Sect. 9.

2 Methodology

We followed a systematic research methodology for the review of literature as described by Tranfield et al. (2003), using a three-stage protocol. In the first stage, we conducted an online search to identify relevant journal papers from the databases. Also, we searched for the articles published between the years 2000 to 2021. Further, the following criterion were used to search for articles related to taxi services and revenue management: ((“taxi” OR “taxi services” OR “aggregator taxi” OR “platform taxi” OR “on-demand transport”) AND (“revenue management”)). The research articles were searched in the following databases, viz.—Springer, Emerald, Taylor and Francis, Wiley, Science Direct, Sage, and Mendeley, using the search criteria as mentioned earlier. Further, the mentioned databases were selected based on their usage in the past by several researchers (Cerchione and Esposito 2016; Ganesh et al. 2018; Baliga et al. 2020), and they provide a complete picture of reputed business and management journals. The primary search resulted in a total of 1594 articles.

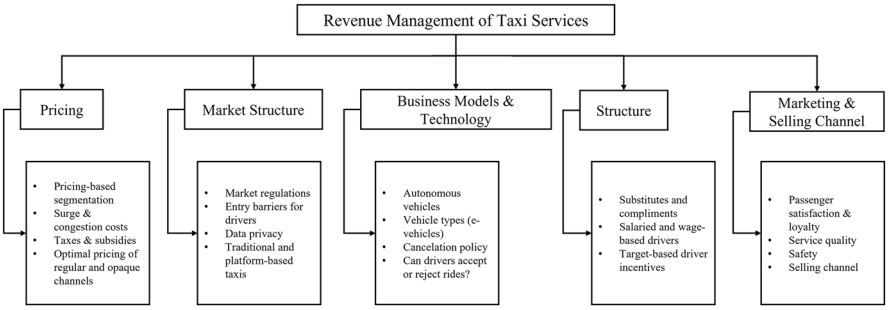

Fig. 1 Revenue management framework for taxi services

Duplicate articles were found and eliminated, and this resulted in a total of 1396 papers. We excluded papers are not a part of Australian Business Deans Council (ABDC)/ Academic Journal Guide (AJG) list. Moreover, we retained the conference proceedings relevant to our study published in reputed conferences. However, we excluded the book chapters, and this returned a total of 197 articles. The famous work of Zorzini on literature review states that an organic search has to supplement the mechanical search to obtain a more comprehensive paper output (Zorzini et al. 2015). The organic search added 19 more articles resulting in a total of 216. We then read the abstract of all the 216 papers, and a final 116 papers were considered for the final study. Figure 2 shows the summary of the methodology used in this paper.

3 Pricing

In this section, we discuss revenue management from the pricing viewpoint. Platform-based taxi services apply pricing strategies such as pricing-based segmentation, surge pricing, taxes and subsidies, and optimal pricing to allocate the right taxi ride to the right passenger at the right price at the right time (Ivanov and Zhechev 2012; Wirtz et al. 2003). Further, we discuss the above mentioned pricing strategies in detail. Traditional taxi services have been operating on a linear fare structure for a very long time where the price paid by the passenger is directly proportional to the distance traveled (Geraghty and Johnson 1997; Yang et al. 2010; Wang et al. 2016). Naturally, drivers prefer long-distance rides since the linear pricing mechanism makes is it more profitable due to economies of scale. But, passengers traveling longer distances are affected by the linear pricing mechanism since it is expensive, and therefore passengers generally avoid using taxi services for long-distance travel.

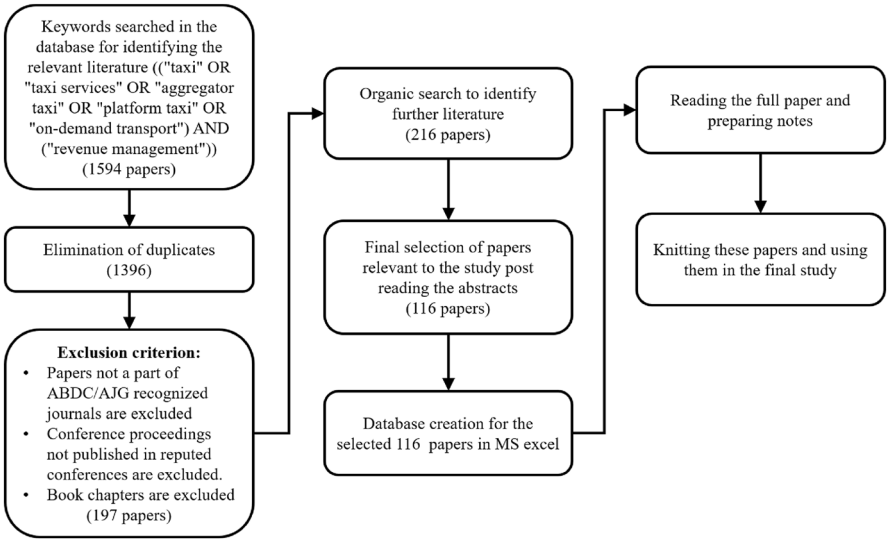

Fig. 2 Summary of literature research methodology

Hence, non-linear pricing models, where the price per unit distance (say kilometer) for a short distance is higher and vice-versa, were developed (Yang et al. 2010). The analytical models developed on differential pricing provide essential insights to several pricing schemes (Cannon and Morgan 1990). The steps for differential pricing include target market selection, dividing the target market into smaller passenger segments, forecasting passenger demand for each segment, computing reservation prices, and calculating each segment's optimal price (Carroll and Grimes 1995; Anderson and Xie 2014).

3.1 Pricing based segmentation

In a taxi market, the unallocated taxis during the low demand times cannot be stored in the inventory for use at the time of high demand. Hence, achieving maximum utilization with a high-profit margin and minimum salvage is vital for taxi services. Several taxi services use the differential pricing strategy to divide passengers into narrow segments. When a taxi company is adjusting its taxi fares based on passengers, time, location, or the type of service, it is said to follow a pricing-based segmentation or differential pricing. For example, a company could differentiate passengers based on the type, viz.—business or leisure. Moreover, taxi services can offer differential pricing based on time; for example, the pricing of a taxi ride at 6 P.M. would be different at 11 P.M. Furthermore, taxi services need not necessarily compute the fare of the taxi ride based on the cost incurred for the service, but the pricing mechanism can be based on the willingness of the passenger to pay for the service (Frost et al. 2018).

The pricing of taxi services is mainly dependent on the sensitivity of passengers. The price-insensitive passengers are primarily business travelers, who plan their travel at the last minute (Yelkur and DaCosta 2001; Lii and Sy 2009). Further, the price-insensitive passengers value their time more, and therefore they are prepared to pay a slightly higher fare. Hence, a price-insensitive passenger is buying a branded taxi service without any discount. On the other hand, the price-sensitive passengers are leisure travelers, who are flexible with pick up and drop times. Also, they keep looking at the competitors to maximize their utility. Therefore, price-sensitive passengers are trying to get a taxi service at a reduced price, and they are not particular about the type of service offered to them. Also, they are prepared to wait for longer times compared to the price-insensitive passengers (Anderson and Xie 2014; Etzion et al. 2006). The pricing strategies based on passenger segmentation can improve the taxi utilization rate, maximize revenues and profits for taxi services. But, it could cannibalize the demand for price-insensitive passengers. By setting optimal prices for all the passenger segments, the taxi services can avoid diluting or cannibalizing the demand for price-insensitive passengers (Rammohan et al. 2022).

3.2 Surge pricing and congestion costs

The spatio-temporal mismatch between passengers and drivers exists due to the non-stationary nature of taxi services. Hence, regulatory mechanisms are required for the

platform-based taxis to control the supply–demand mismatch struggle during peak hours (when demand is high) and off-peak hours (supply is high). Hence, to maintain market efficiency, Uber introduced a surge pricing mechanism (Qian and Ukkusuri 2017). The surge price is the multiplication of standard fare and the surge multiplier. Taxi fares during the peak hours are updated based on the real-time demand, and the surge can change rapidly. Also, surge pricing varies across different areas, and therefore some areas might have a higher price depending on the demand for taxi rides (Uber-Technologies 2022). More drivers are encouraged to enter the taxi market at times when the surge multiplier is high since they benefit from the transient spike in taxi fare (Harding et al. 2016).

Platform-based taxi services connect passengers and drivers at times even when the distance between them is very long, which typically occurs when the taxi companies are excessively burdened. As a result, drivers take up the wild goose chase for providing service to distant passengers. Further, the taxi utilization rate comes down due to wild goose chase, resulting in a reduced profit margin for the drivers (Castillo et al. 2017). As a result of wild goose chase, the two-sided digital platforms are susceptible to market failure compared to traditional taxi services (Arnott 1996).

The platform-based taxi services use surge pricing mechanisms to minimize the risk of wild goose chase (Castillo et al. 2017). But, practitioners and researchers agree that surge pricing is not the only way to tackle wild goose chase. Strategies like limiting the number of rides during the peak demand, fixing a territory for drivers, and allowing drivers to accept or reject rides are other ways of handling wild goose chase (Afeche et al. 2018). However, both these strategies compromise the advantage of two-sided digital platforms over conventional taxis.

Since the demand for taxis is high in most urban cities, traffic congestion is inevitable, especially when the road resources are limited. Therefore, platform-based taxi services must regulate the congestion externalities of vacant and non-vacant taxis to maximize the taxi market efficiency. Also, taxi services can promote ride-sharing mechanisms to enhance operational efficiency (Lin et al. 2012).

Few drivers avoid taxi rides since congestion costs are not included in the overall pricing scheme. The linear fare structure of taxi services may not involve the congestion costs, and as a result, the driver's opportunity cost tends to be higher than the linear fare. Hence, some drivers will choose not to work during peak hours which will hamper the service quality of the taxi company. Incentivizing taxi drivers during peak hours is a strategy implemented by some e-hailing taxi companies (Gan et al. 2013). Congestion pricing is a phenomenon that can affect the urban system, and neglecting the effects could lead to an irrational pricing policy. The impact of congestion price can directly affect the transport systems, travel patterns, and overall urban spatial structure (Zhong et al. 2015).

3.3 Taxes and subsidies

Demand for taxis in urban areas keeps rising due to the accrued city population and diverse social enterprises. Hence, urban areas face traffic jams, and drivers are reluctant to accept taxi rides to congested areas. As a result, passengers are offered

discounts for taking rides to high-demand locations and are charged higher fares when taking rides to low-demand places (Zeng and Oren 2014). On the other hand, offering subsidies to drivers for taking rides to congested areas will ensure higher service levels to passengers (Pueboobpaphan et al. 2019). Hence, the topic of charging a higher fare to passengers or providing subsidy to drivers for taking a ride to low demand area is debatable.

Most drivers generally tend to cruise towards high-demand areas such as the city center, airport, railway station for finding passengers. As a result, low-demand areas are affected by the shortage of taxi supply. Therefore, levying taxes for taking rides from high-demand areas and offering subsidies for taking rides in low-demand areas could be a strategy to maximize social welfare. Subsidy programs have eased the difficulty of taking taxis in the urban market, especially during peak hours. Companies such as Didi and Kuaidi have spent billions of dollars in subsidies for getting market share (Wen et al. 2017). Few studies on the effects of offering subsidies identify that it is profitable to provide subsidies to drivers compared to giving discounts to passengers. (Su et al. 2018). However, platforms must understand that subsidy programs result in a rapid increase in the number of short-distance rides of the drivers (Leng et al. 2015). Hence, designing optimal subsidy programs for platforms is crucial for profitability and social welfare. So, the idea of subjecting platform-based taxis to the same taxes and subsidies of traditional taxi services is an interesting topic for debate.

3.4 Optimal pricing

Even today, few taxi companies utilize linear pricing mechanisms for calculating the passenger fare. As a result, long-distance travel becomes expensive for passengers, and hence some might opt for unauthorized taxi services, which may not be safe. Therefore, a non-linear pricing mechanism was recommended by Hong Kong's Taxi Advisory Committee that charges higher fares for short-distance rides and lower fares for long-distance travel. Yang et al. (2010). Pricing of taxi services depend on several factors, viz.—sales channel, pricing mechanism. Selling channels could be regular or opaque, while the pricing mechanism could be data driven or market based. Under the topic of optimal pricing, we discuss: regular and opaque channel pricing, and pricing mechanism.

3.4.1 Regular and opaque channel pricing

Passengers can purchase a branded service in the regular channel with complete information and no additional waiting times by paying a premium. On the other hand, opaque channels sell taxi rides at a reduced price to passengers. Nevertheless, a passenger buying rides on opaque channel does not have complete information due to opacity, i.e., certain features like the type of vehicle, GPS tracking, driver ratings are not available to passengers before the sale of rides. Further, there is additional waiting time associated with opaque channels and the passenger will not know the type of car (sedan, hatchback or SUV) before the sale of rides. The role of opaque

channel is to segment passengers and offer taxi services at a discounted price without cannibalizing the demand in the regular channel. So, the key factors that differentiate regular and opaque channel pricing are: opacity and waiting costs (Etzion et al. 2006; Anderson and Xie 2014; Hummel 2015; Feng et al. 2018). Using regular and opaque channels simultaneously has gained a lot of attention in taxi services. However, there is a possibility of the opaque channel cannibalizing the demand of regular channel.

3.4.2 Pricing mechanisms

Two-sided markets typically use data-driven methods for pricing taxi rides in the regular and shared channels. Generally, taxi platforms use data-driven approaches for computing the taxi fare. But, few companies like GrabTaxi and Liftago have started moving towards a market-based approach by incorporating auctions for pricing taxi services, and the market-based mechanism has proven to be a viable option in terms of financial returns (Zhou and Xu 2017; Egan et al. 2019). Taxi platforms are moving towards dual-channel sales, where both data-driven and market mechanism-based pricing methods are used. However, researchers and practitioners agree that calculating optimal fares separately in the dual-channel may not be optimal (Hummel 2015; Feng et al. 2018). Hence, joint optimization of prices in the dual-channel is an essential strategy for platforms to maximizing revenue (Etzion et al. 2006).

Pricing mechanisms for taxi services are shown in Fig. 3. In the market mechanism, passengers pay the fare computed by the platform, and drivers pay a convenience fee to the platform. In the market mechanism, passengers quote the maximum price they are willing to pay for the ride. Any quote that exceeds the secret reserve price (computed by the platform) wins the auction, and the passenger will get a taxi service from the platform. Likewise, drivers quote the minimum price with which they can offer the taxi ride. Any quote below the secret reserve price (set by the platform) will win the auction, and the driver can offer the service. Here, the challenge lies in the computation of secret reserve prices for passengers and drivers.

4 Market structure

In the market structure aspect of the revenue management, we consider the two critical elements, viz.—market regulations and technology. In most geographies, taxi markets are regulated in terms of pricing, fleet size, operations, safety, etc. Also, the market structure can be classified based on the technology used, viz.—radio or app-based. The technology-based transport model, viz.: platform-based taxi services have opened a new taxi market. Nevertheless, the applicability of road-hailing taxi services' control and regulatory aspects to platform-based models is debatable (Barglind 2015). The platform-based taxi services expand business by entering new geographies, and the debate over platform companies as "technology-middleman" or "unregulated taxi service" is still ongoing.

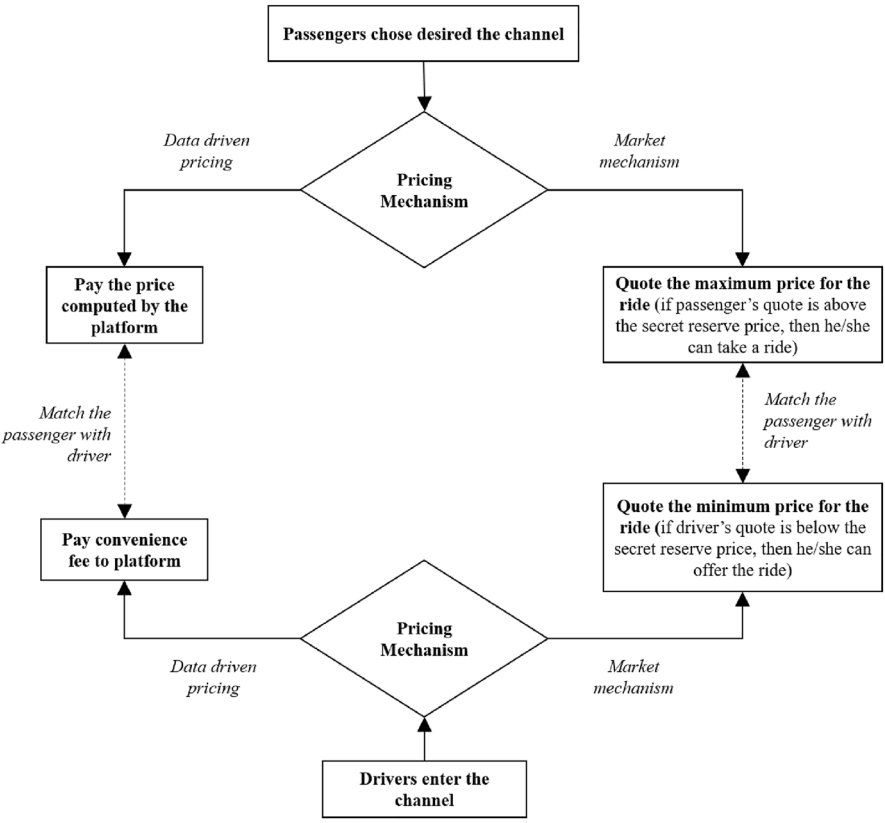

Fig. 3 Pricing mechanism for taxi services

4.1 Market regulations

Taxi regulations can be summarized under two aspects, viz.—economic and non-economic. The economic angle focuses on price and entry controls, while the non-economic angle pertains to quality and safety standards. Most of the taxi markets compute fares and impose regulations concerning vehicle maintenance and operations (Harris 2002). Further, most taxi market regulators control, viz.—new entrants in the taxi market, pricing decisions, driver eligibility, vehicle status, financial and operational requirements.

By regulating taxi services and enforcing contractual relationships, the digital platforms can provide economical benefits for socially marginalized groups, viz.—unemployed and refugees (De Stefano 2015; Byrne and Waters 2015). Although platform taxis bring in numerous benefits such as providing flexibility and increase competition, they can negatively affect drivers due to long-term employment benefits, irregular incomes, inconsistent scheduling, exclusion from corporate benefits, and protection. Further, digital platforms have the disadvantages such as experience

absenteeism and poor productivity from drivers (Johnston and Land-Kazlauskas 2018). Thus, companies like Uber have started to enact a neoliberal playbook that provides individual protections of property rights, freely operating markets, equitable trade, and a "legal framework of freely negotiated contractual obligations between juridical individuals in the marketplace. The sanctity of contracts and the individual right to freedom of action, expression, and choice must be protected... at all costs" (Zwick 2018).

4.2 Entry barriers for drivers

The New York City Taxi app imposes a condition on the driver's age to be at least 19, while Uber and Lyft require a driver to be 21 years old. Other requirements for the taxi drivers are, viz.—driving license, social security number, driving record history, and background check. The proportion of female drivers are constantly increasing in the platform-based taxi services, and studies demonstrate that female drivers are more educated than their male counterparts (Yaraghi and Ravi 2017).

In the non-digital platforms, drivers are often treated as salaried employees and are required to qualify themselves by obtaining a driving license or "medallion" for offering transport services. The regulations vary between geographies. For example, the passenger pick up may be limited to the jurisdiction defined by the medallion (Wang and Yang 2019). Regulatory bodies typically restrict pricing and fleet size. A considerable amount of research has been done on developing taxi-based economic models on the governmental regulations (Douglas 1972).

4.3 Data privacy

Personal privacy protection has become a cause for concern due the recent technological developments. In order to optimize traffic management applications, viz.—urban planning, traffic controlling, route recommendation, taxi services collect precise vehicle trajectory data (Zheng et al. 2011). By collecting the GPS data, taxi services have access to individual's trajectory, and can therefore identify the individual's travel pattern. Along with the passengers travel behavior, the driver's personal habits can also be exposed (Sui et al. 2014; Douriez et al. 2016). Thus, taxi services must anonymize the sensitive information that can avoid jeopardizing privacy of passengers and drivers.

The taxi platforms collect a huge amount of passenger data to infer the purchasing behavior of passengers. Hence, a passenger must restrict access to their information and place regulations on the collected data. Also, taxi platforms identifying patterns in passenger purchases and information like passenger's home and workplace are available in the application history. Information like passenger's highest purchase price and willingness to pay can increase the efficiency of the platform operations but can sometimes lead to price discrimination against passengers (Yaraghi and Ravi 2017).

4.4 Traditional and platform-based taxis

The road-hailing traditional taxis are utilized as supplements and compliments for public transport services. But, they create congestion and pollution-related problems when they move around in search of road-hailing passengers. Also, the matching efficiency concerning time and space is low (Wang and Yang 2019). Moreover, the inefficient design of taxi bays in handling capacity at high-demand areas such as airports leads to long queues (Wong et al. 2015). These problems can be effectively managed by technology compared to regulatory policies (Manski and Wright 1967).

In the traditional road-hailing taxi models, the taxi utilization rate is much lower compared to the e-hailing taxis due to information asymmetry and spatial differences. Also, the conventional pricing assumes a correlation between variables, viz.—both endogenous and exogenous- which can be seen in Fig. 4. The passenger searching behavior of vacant taxi drivers is modeled using multinomial logit models. Typically, the multinomial logit models are calibrated and validated using GPS data (Sirisoma et al. 2010; Wong et al. 2014). From Fig. 5, we can infer that the fleet size positively affects the number of vacant taxis, i.e., the more the fleet size, the more vacant taxis.

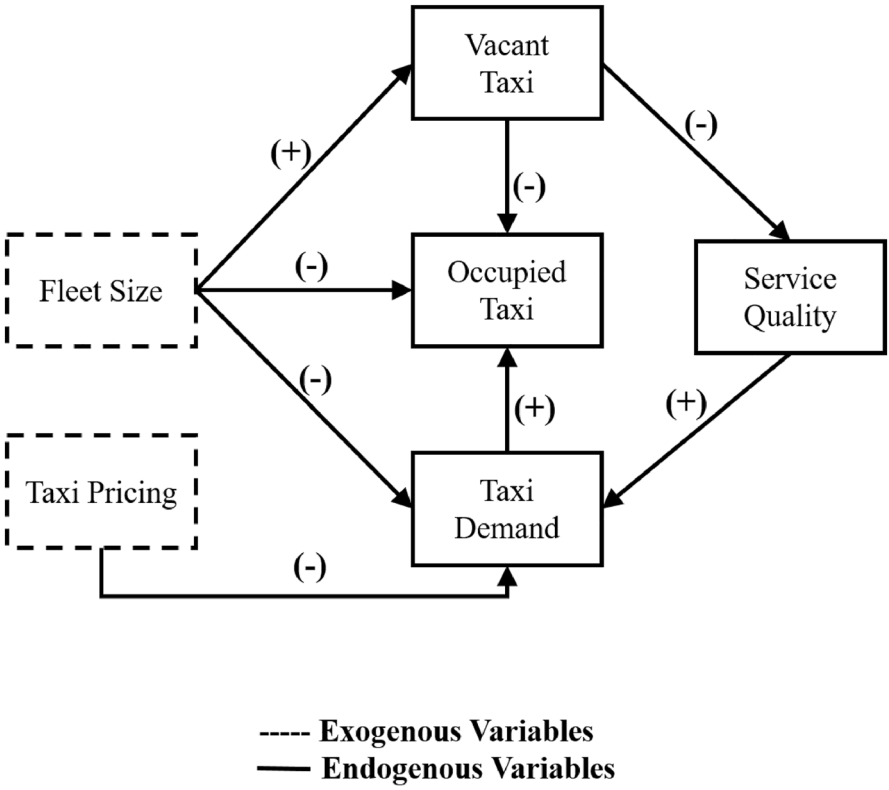

Fig. 4 Assumed correlations in road-hailing taxis

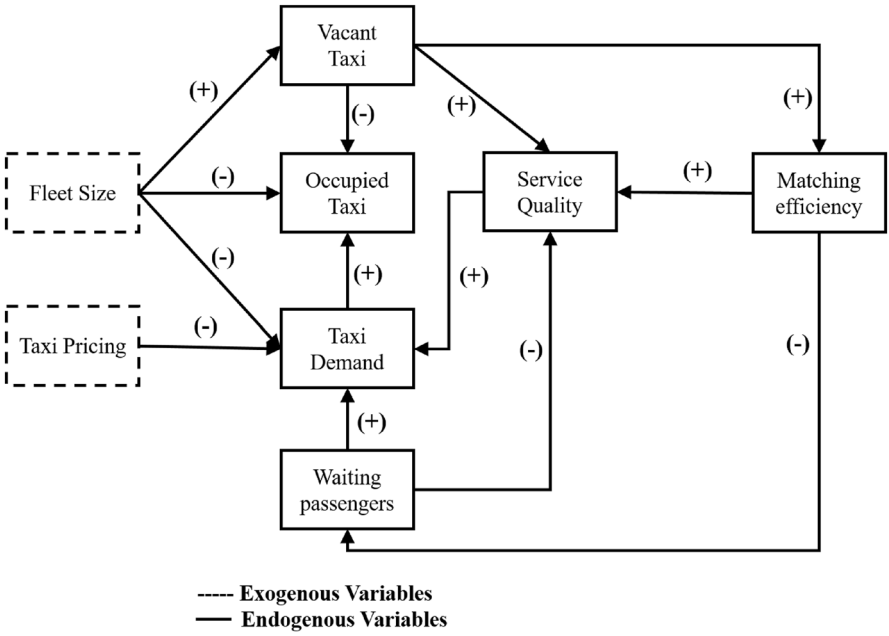

Fig. 5 Assumed correlations in e-hailing taxis

Also, the more the number of vacant taxis, the higher are service quality and matching efficiency. Further, the significant distinction between road and e-hailing taxis is the matching efficiency. As the matching efficiency increases, the number of vacant taxis reduces, the service quality increases, and passenger waiting times decrease. Thus, increasing the matching efficiency by developing algorithms is vital for e-hailing taxi services to increase profits and achieve market share.

4.4.1 Platform-based taxi services

Platform-based taxi services can be categorized into two, viz.- two-sided and one-sided platforms. Two-sided platforms act as intermediaries and allow the affiliated passengers to connect directly with affiliated drivers (Hagiu 2007). Further, it connects passengers with the nearby taxis, and the drivers are allowed to accept or reject the requests. On the other hand, one-sided platforms, also called as taxi dispatching centers, will assign the taxis to passengers based on a matching algorithm, and the drivers are treated as salaried employees. Although this service model provides a higher successful matching ratio, the freedom of accepting or rejecting taxi rides is compromised (Wang et al. 2016).

5 Business models and technology

Business models are linked with technological advancements and they help companies to organize and sell products or services differently (Baden-Fuller and Haefliger 2013). Moreover, the technological advancements help companies to cater to the needs of an emerging market requirement (Kavadias et al. 2016). Taxi markets have experienced different business models over time. For example, Uber disrupted the taxi market with an unconventional business model, where they neither owned taxis nor employed drivers. Hence, it is often referred to as a technological company since it matches passengers and drivers through a mobile application (Walji and Walji 2016). Also, we see autonomous vehicles as a vital factor for the platform-based taxis for increasing their profits. In addition to the technological front, the vehicle types, riding channels, and cancellation policies are essential aspects of revenue management that require taxi services' attention.

5.1 Autonomous vehicles

Taxi services view autonomous vehicles as another technological disruption that can change platform economics. Also, autonomous vehicles could be a solution for reducing the number of accidents caused by human errors (Kaur and Rampersad 2018). The public trust is a major barrier in adapting to autonomous vehicles (Bansal et al. 2016). Moreover, the significant chunk of platform revenue goes to drivers as wage payment, and the introduction of autonomous vehicles can eliminate the variable cost of driver payments (Siddiq and Taylor 2019). Autonomous vehicles can minimize the parking space requirements and considerably reduce the taxi fleet size (Stoiber et al. 2019).

Currently, taxi services are working towards the sharing of driving tasks between human drivers and the vehicle. The *handover/takeover* transition systems, i.e., a human driver passing control to the car and vice-versa is crucial for achieving the intermediate level of automation (Majzik et al. 2019). With regard to human–machine interactions, taxi services must address the research gaps in the areas of risk reduction, legal systems development, and ethical issues handling (Bellet et al. 2019). In this regard, interactions between autonomous vehicles and the environment using techniques like reinforcement learning have gained much attention. Furthermore, designing a framework for the driving policy of autonomous vehicles using artificial intelligence requires more research and development.

5.2 e-vehicles

Air pollution in metropolitan cities has become a critical issue, and studies show that road transport is the most vital contributor to air pollution (Kim et al. 2017). Electric vehicles or e-vehicles are viewed as a sustainable solution to prevent roadside pollution and greenhouse gases (Nanaki and Koroneos 2013). In urban areas, taxi drivers take up short-distance rides but do high yearly mileage. The

stop-and-go feature of e-vehicles improves the vehicle's efficiency, especially during traffic congestion. Combining high yearly mileage and low running costs makes e-vehicles a sustainable alternative for taxis running on conventional fuel. Further, e-vehicles provide a comfortable ride to passengers and drivers with reduced noise and vibrations. However, e-vehicles require frequent charging that makes it challenging in specific geographies due to the unavailability of charging infrastructure. Also, from a socio-economic point of view, e-vehicles do not create roadside pollution and traffic noise and contribute to high service quality (Scorrano et al. 2020; Zhou et al. 2021). Few studies on platform taxis demonstrate that transition into e-vehicles could save fuel and reduce operational costs. Nevertheless, research on platform taxis transitioning into e-vehicles is still at the preliminary level (Carpenter et al. 2014).

In China, the government plans to move the taxi fleet to e-vehicles, especially after the successful deployment of e-taxis in the cities of Taiyuan and Shenzhen. Further, cities like New York and London promote e-taxis and are slowly moving towards electrification (Parmentier et al. 2021; Scorrano et al. 2020). However, the crucial factor for the industrialization of e-vehicles is the network design for battery charging stations. Hence, there is a need for location optimization models. Moreover, developing queuing models for battery charging is another vital area for managing demand and supply. Further, pricing models for vehicle charging are another area of focus for the successful implantation of e-vehicles.

5.3 Cancellation policy

Business domains such as hotels, airlines, and taxis are prone to cancellations that create another opportunity to increase revenue. During cancellations, the passenger refunds are time and segment-based. The penalty for no-shows is different from cancellations. Moreover, overbooking is another area for revenue generation, and organizations use a penalty-cost function for determining the penalties (Subramanian et al. 1999; Talluri and Van Ryzin 2004; Sierag et al. 2015). The penalty-compensation strategy is vital for platforms since passengers are prone to frequent cancellation of taxis, especially when they meet a road-hailing taxi. Such frequent cancellations may decrease the driver's taxi utilization rate. Hence, an appropriate penalty strategy is required to balance the taxi supply in the market (He et al. 2018).

Traditional taxis treat drivers as salaried employees and do not have the freedom to accept or reject taxi rides. In contrast, the platform-based taxis offer flexibility to drivers by capturing their preference for accepting and rejecting taxi rides. However, traditional taxi services provide higher matching efficiency compared to platform taxi services. In the case of platform-based taxi services, employment issues may arise. Focus on the employment issues of drivers, i.e., treating them as employees or independent contractors is crucial. In the traditional taxi services, a manager is responsible for: enacting discipline, termination of drivers from the company, approving bonus. But, in the online platforms, the computer algorithm plays the managerial role that may not understand the driver emotions (Zwick 2018).

6 Structure

The identified structural elements and characteristics help to make sense of revenue management literature in the developing areas. In the structural aspects of the proposed revenue management framework, we discuss: substitutes and complements, salaried and wage-based drivers, and driver incentives.

6.1 Substitutes and complements

Prolific advancements in the shared transportation have transformed taxi services to complete mobility solution providers. For example, Uber offers a plethora of services, viz.- UberX (low-priced service), Uber Black (high-priced executive service), Uber Pool (Shared rides), Taxi (regular platform taxi service), SUV (premium service with additional seating). Didi, the largest platform taxi company in China, offers services such as Taxi (regular platform taxi service), Express (low-priced service), Premier (medium-priced service), Luxe (high-priced executive service), Pool (shared rides), Designated Driving (designates chauffeur for vehicles owned by customers), Bus and many more (Wang and Yang 2019). Although offering a variety of services helps taxi services generate revenues from different segments of passengers, the low-priced channels might cannibalize the higher-priced channels' demand. Hence, taxi services must devise mechanisms to avoid the cannibalizing effect of one channel over the other.

6.2 Salaried and wage-based drivers

Platform-based taxis provide flexible and impermanent opportunities, unlike standard full-time employment that provides permanent employment. As discussed in Sect. 4.4.1, one-sided platforms generally offer salaried employment, and two-sided platforms operate on wage-based payments. The major difference between the one-sided and two-sided platforms is the volatility of the incomes. Although the volatility of income generated by drivers in the one-sided platforms is low, the risk of losing jobs is high, especially during financially challenging times (Green 2014).

6.3 Driver incentives

In geographies like India, taxi drivers are offered lumpsum incentives in addition to the taxi fare for increasing the supply of taxis. Typically, incentives given to the drivers are trip-based or earnings-based (Ayyar 2017; Kashyap and Bhatia 2018). In the trip-based incentive scheme, drivers must cross the target threshold that is set by the platform, say ten trips on a given day or seventy trips in a week. Likewise, in the earnings-based incentive scheme, a driver is expected to cross the target threshold, say the driver's earnings must exceed the predefined target on a given day. After crossing the

target threshold (in both trip and earnings-based), the drivers are given a fixed lump-sum incentive.

When platform-based taxi services started operations, they offered generous incentives to taxi drivers. With time, platform aggregators modified the incentive policy by setting challenging target thresholds and reducing the quantum of incentives that affected drivers' earnings. In the year 2017, drivers take-home earnings declined by 33% due to the drop in the quantum of incentives (Variyar 2018; Ayyar 2017). So, designing optimal incentive schemes for drivers are crucial for ensuring seamless supply of taxis in the market.

7 Marketing

In the marketing view point of revenue management, we discuss service quality, safety and selling channels. Satisfaction in the context of taxi services can be defined as the inclusive attitude of passengers with respect to the platform aggregator. Passenger satisfaction could be treated as an emotional reaction to the difference between the expected and obtained services (Hansemark and Albinsson 2004). A satisfied passenger is loyal to the platform and makes repeat bookings, and promotes the brand through word of mouth (Angelova and Zekiri 2011). In the context of taxi services, factors such as driver behavior, service quality, accurate billing, passenger waiting time, and perception of safety by passengers influence passenger satisfaction (Horsu and Yeboah 2015).

Platform-based taxi services entered the market with the objective of capturing new passengers by offering taxi rides at a competitive price. Due to market saturation, their current focus is primarily on passenger retention (Nguyen-Phuoc et al. 2021). Taxi services see loyalty and customer satisfaction as key factors for achieving long-term financial returns (Nguyen-Phuoc et al. 2020). Developing loyalty ensures a competitive advantage and facilitates maintaining the market share of platform taxis. Moreover, marketing research in the field of platforms emphasizes more on understanding the factors influencing passenger satisfaction and loyalty (Lam et al. 2004). Free trials, promo codes, referral bonuses, and discounts during non-peak hours are provided to passengers to keep the existing pool of passengers away from competitors.

Further, platform taxis offer lumpsum incentives to drivers in some geographies like India to ensure driver loyalty. The practice of incentivizing drivers ensures a seamless supply of taxis to the passengers. The driver incentives are of two types, viz. - ride-based and cost-based. The ride-based incentives are provided for completing a certain number of rides in a given timeframe, while the cost-based incentives are given for taking rides until specified earnings are made by the driver in a given timeframe (Rammohan et al. 2022).

7.1 Service quality

Service quality is a measure of efficiently delivering a service concerning the passenger's expectations. Public transport in specific geographies is characterized by long waiting times, overcrowding, uncertain travel times, undependable services

(Adarkwa 1991). One of the overlooked aspects of commuting is service quality. Also, the primary concerns for passengers with respect to public transport are, viz.—affordability, reliability, waiting times, comfort, driver behavior, and continuous service (Abane 1993). Further, factors such as: ventilation, cleanliness, safety are some of the concerns related to service quality (Jorgensen et al. 1999). Studies indicate that waiting and travel time, safety, comfort, driver behavior, reliability, and continuous service are the strongest determinants of passengers perception of service quality compared to the fare (Alonso et al. 2018; Askari et al. 2021).

7.2 Safety

Researchers in the field of taxi services unanimously agree that safety is one of the biggest concerns in choosing a particular taxi service (Nguyen-Phuoc et al. 2021). The literature on taxi services agrees that platform-based taxi services provide a higher level of safety for passengers in comparison with traditional taxi services. The availability of GPS-enabled mobile devices provides accurate location information and a sense of safety to the passengers. Further, passengers have the complete details of the driver, viz. - driver name, vehicle registration number, driver rating, driver photo, before the sale of the taxi ride (Shah 2020). The driving behavior of taxi drivers is influenced by ride-based income and incentives leading to quick transport of passengers to their destination. Hence, taxi drivers require training and support to ensure their personal and passenger safety (Aghabayk et al. 2021).

7.3 Selling channels

Selling the rides through appropriate channels is crucial for platform-based taxi services. Due to the recent advancements in smartphone and telecommunication technologies, the platform-based taxis can offer the sale of taxi rides through multiple channels. The riding channels can be prepaid, rental, regular or shared. Two most commonly used business models by one-sided platforms are prepaid and rental taxis. Prepaid taxis are common in airports and railway stations, and are governed by regulations with respect to pricing and branding. Typically, taxi rides are sold to walk-in passengers via of counter-based sales. Rental taxis use a radio-based communication technology for matching passengers and drivers.

Platform-based taxis primarily focus on business travelers, where they sell a branded taxi ride with high transparency and no additional waiting times at fixed posted prices. Recently, the platform-based taxis introduced a new channel, viz. - group ride—where passengers with a similar origin–destination and corresponding starting times are grouped into a single ride. The group ride channel has gained high popularity due to its pricing strategy where the rides are sold at a reduced price compared to the regular channel. However, the group ride channel operates on opaque selling strategy where certain features like the type of taxi (sedan or SUV), driver ratings, and GPS tracking are not available to the passengers before the sale of the taxi ride (Rammohan et al. 2022).

Competition between platforms is creating demand for a new segment of passengers who were previously priced out of the market. Most of the current taxi services treat passengers as business travelers and operate on a fixed pricing strategy. A few platform-based taxi services like Grab Taxi and Lyft are moving towards a market-based mechanism where leisure travelers are given an opportunity to enter the platform through the auction channel (Rammohan et al. 2022). Creating a new demand ensures continuous income for the platform aggregator during non-peak hours and increases their respective market presence.

Platform-based taxis come under the broad umbrella of two-sided platforms that connect passengers and drivers via mobile application. Also, they sell taxi rides via two channels, viz. - regular, shared and opaque. Regular rides are the cash cows of taxi services that have been popular among the commuters planning their travel at the last minute. Passengers purchase a branded service with complete information in the regular channel (Anderson and Xie 2014). On the other hand, shared rides have been recently introduced by taxi services, where passengers with common destinations and coinciding starting times are grouped into one ride (Qian et al. 2017). The shared rides are a variant of opaque selling, where the passengers pay a reduced price compared to the regular ride. Due to opaqueness in the shared channel, a passenger will not know the type of car, time to reach the destination, and the number of co-passengers before the sale of rides. Also, there are strict penalties charged for no-shows and cancellation of rides by the passenger. Companies like Grab Taxi and Lyft have personified the usage of auction channel for selling taxi rides. Auctions have proven to be a viable option in terms of financial returns. Moreover, the recent advancements in the telecommunication technology has made the incorporation of auction channels operationally efficient (Zhou and Xu 2017; Egan et al. 2019). Business models of taxi services are shown in Fig. 6.

8 Discussions

Revenue management processes are attractive in the service industry since they establish rational methods to maximize revenues and profits by utilizing differential pricing. Companies ignoring revenue management when their rivals implement are at a disadvantage. Also, “What you don’t know about Revenue Management could kill you” was the reflection of Donald Burr, former CEO of PEOPLExpress (Bryan 1989). Hence, the need for a complete understanding of revenue management and its implementation is vital.

Even when the organization exhibits profitable performance by incorporating revenue management practices, it is critical to understand which system’s feature had a significant share of gain and what more could have been attained. The mechanism that had a strong influence on the performance should be identified and must be replicated. Further, the revenue management system must be capable of tracking the lost opportunities and their impacts. A complexity associated with the pricing models is the number of variables related, viz.—type of vehicle, travel time, sales channel, and surge multiplier. Hence, measuring the performance of revenue management concerning pricing has to be approached in phases. Along with measuring gain

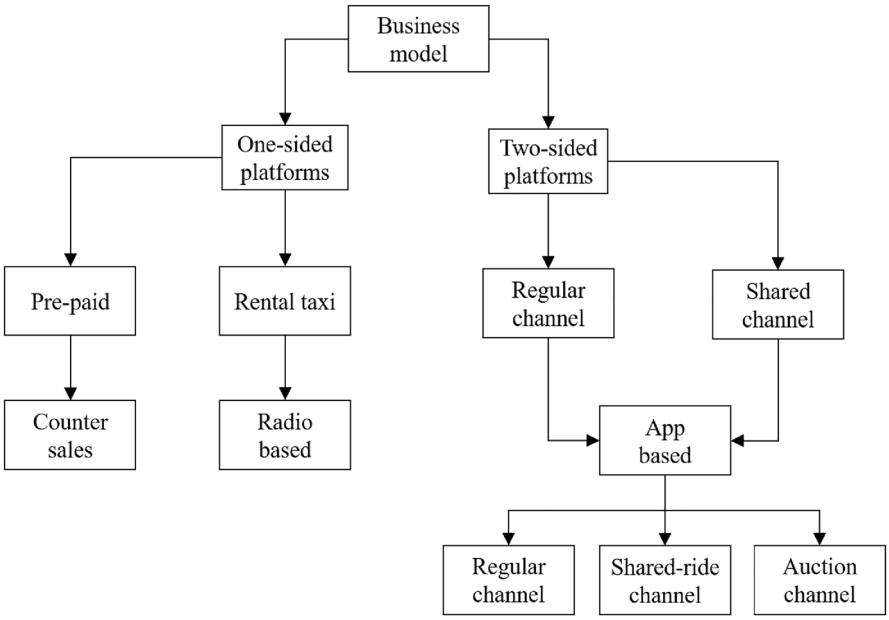

Fig. 6 Business models of taxi services

and calculating value, linking revenue management systems and performance measurement is vital for organizations (Anderson and Blair 2004). Hence, a thorough understanding of the revenue management framework for platform-based taxi services is crucial for managers. This paper provides a conceptual framework from the viewpoints of pricing, market structure, business models of technological advancements, structural and marketing that can help managers make informed decisions with regard to revenue maximization and profits.

8.1 Future research

We find a need for research in dual-channel optimal pricing, incentive design, and alternative pricing mechanisms from the identified literature review. Also, the gaps mentioned above have not been studied from the revenue management perspective in the academic literature. Nevertheless, researchers and practitioners must critically examine the gaps before classifying them as worthy of attention (Nicholson et al. 2018). The future research topics along with the academic and practical implications are discussed in Table 1.

Few companies like IBM are moving towards a dual-channel strategy, where some of its products are sold on e-bay at a reduced price by running auctions, while the same products are sold on the IBM website at a premium price using the posted price mechanism (Etzion et al. 2006). Also, dual channels are proven to be profitable for platforms when they are optimized simultaneously (Chen et al. 2014; Huang et al. 2017; Shapiro

Table 1 Research gap identification

Research gaps	Implications-Academia	Implications-practice
How can platform-based taxi service create additional demand from a new segment of passengers?	Research on creating additional demand is crucial for platform-based taxis to generate additional revenue and keep the drivers engaged during the non-peak hours	Platform-based taxis must experiment incorporating new selling channels such as group rides and auction channels to ensure higher financial returns
What are the challenges in implementing a dual-channel strategy?	Simultaneous optimization of regular and opaque channels is vital for the platform's revenue maximization. To the best of our knowledge, optimizing a dual-channel in the presence of three agents is not available in the literature	Platforms can avoid the effect of opaque channels cannibalizing the demand of the regular channel by setting optimal prices in regular and opaque channels concurrently
Why should taxi platforms provide driver incentives?	To the best of our knowledge, no literature is available on driver incentive design	Understanding the driver's behavior with regard to incentives is essential for platforms to ensure a seamless supply of taxis in the market
What pricing schemes are acceptable to passengers and drivers?	Mechanisms that capture the passenger and driver preferences are in the preliminary stages of development. Also, pricing-based passenger segmentation can be achieved by moving from linear pricing to auction-based models	Pricing mechanisms like auctions have proven to be a viable option in terms of financial returns, and taxi platforms are still in the preliminary stages of implementing auctions for pricing taxi rides
What are the challenges in implementing of e-vehicles?	Location optimization, queuing models, and pricing models for vehicle charging are essential areas of research and development to implement e-vehicles successfully	Promoting e-vehicles by offering driver subsidies could increase the share of e-vehicles
What kind of business models and technological advancements are required for launching driver-less taxis?	Modeling the interaction between the autonomous vehicles and the environment, determining optimal driving strategy are vital for launching autonomous vehicles	Proper planning and implementation of autonomous taxis can save millions of dollars for platform-based taxis
Competition between platform-based taxis	Modeling passenger and driver behavior under network congestion, incentives in the presence of multiple platforms require further research	Managers and practitioners can improve market share by understanding the customer and driver behavior in more detail

2011). To the best of our knowledge, dual-channel optimization models for taxi services have not been explored. Furthermore, optimizing dual-channel in the presence of three agents has not been studied in the academic literature. By optimally pricing the rides in the dual-channel, taxi services can segment passengers and generate additional revenue without cannibalizing the demand of the regular channel.

In addition to the trip fare, offering lumpsum incentives to drivers for increasing the supply of taxis is practiced in some geographies like India. However, the drivers' earnings have declined drastically since the platforms have reduced the quantum of incentives with time (Ayyar 2017; Variyar 2018). Moreover, due to the poor demand during peak hours, drivers face difficulty reaching the threshold number of rides to achieve the incentives. Therefore, platforms must identify strategies for increasing the demand during non-peak hours. To date, there is no literature available on the platforms incentive policy framework. Also, understanding drivers' behavior with regard to incentives is vital for the platforms to ensure a seamless supply of taxis.

From the review, we identify the need to study the differential pricing schemes of taxi services. Platform taxis reduce the spatial barrier between passengers and drivers by using a mobile application. However, the passengers and drivers are left with no choice but to accept or reject the ride. Hence, there is a need for developing mechanisms that capture the preference of passengers and drivers. Taxi platforms such as Didi Dache have operated on a linear pricing mechanism, where the fare paid by passengers is proportional to the distance traveled (Zhou and Xu 2017). Due to the linear pricing mechanism, passengers avoid long-distance taxi rides as it tends to be expensive. Therefore, developing mechanisms that capture the preference of passengers and drivers can increase the share of long-distance rides (Yang et al. 2010). Few companies such as Liftago and GrabTaxi are moving towards a market-based mechanism by using auctions to price taxi rides. Also, auctions have proven to be viable in terms of financial returns (Zhou and Xu 2017; Egan et al. 2019).

New business models are essential factors for the success of platform taxis that can improve their revenue and protect the market share against the competitors. However, implementing business models such as e-vehicles and autonomous vehicles requires a lot of study regarding planning and implementation. Promoting e-vehicles could reduce roadside pollution drastically, but the infrastructure for battery charging stations requires further investigation. Also, studying queuing models in the charging stations are essential for minimizing the queue length and waiting times of customers. The pricing schemes for charging station is another area of study identified from the review. Likewise, the usage of autonomous vehicles could save millions of dollars for platforms. But, there is a need for further research in developing a theoretical framework for the driving policy of autonomous vehicles. Furthermore, modeling the interaction between autonomous vehicles and the environment requires attention from academia.

9 Conclusion

It is evident from the increasing number of publications over the last twenty years that the research and development in the revenue management of taxi services have attracted significant attention from researchers and practitioners. This review

focuses on four critical aspects of taxi services' revenue management practices: pricing, market structure, business models, and structural elements. The pricing strategies of taxi services are moving towards a market-based mechanism that can capture the preference of passengers and drivers. Moreover, incentives offered to the driver will play a crucial role in ensuring a seamless supply of taxis in the market. In the future, taxi business models are moving towards a dual-channel mechanism, where the platform will simultaneously sell rides using posted price and auction mechanisms. In addition, the governmental authorities will actively promote the e-taxi business model by offering generous subsidies to drivers to purchase and maintain e-taxis. Furthermore, the autonomous vehicle business model has the potential to save millions of dollars for taxi platforms. The war between taxi platforms for market share will increase the service quality, especially the taxi fare and waiting time of passengers will further go down. With the introduction of new technologies, viz.—autonomous vehicles and auction-based pricing, the performance monitoring of revenue management systems will be the key for taxi services to track the growth and future potential.

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Conflict of interest The authors declare that they have no conflict of interest.

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